5.4.5 Choosing a knowledge graph service

As the complexity of your digital twin grows, you may no longer want to manage your own graph database infrastructure to host your digital twin's knowledge graph. At this point you can make use of managed services within public cloud providers to host it for you. Both AWS and Microsoft offer fully managed services for the developing knowledge graphs for digital twins. I trialed building my home graph in both Azure Digital Twins and AWS IoT TwinMaker, but ultimately decided on Memgraph for it's deployment flexibility—it runs seamlessly as a Docker container both locally for development and on AWS for production, enabling rapid experimentation without infrastructure constraints. This portability allows me, and you the reader, to quickly prototype and iterate on different graph models making it ideal for the exploratory nature of home digital twin development. As the database becomes larger and more complex and I look to a managed service, I can migrate my data to one of these cloud services, whilst maintaining the same graph traversal semantics I use today.

5.5 Standards and interoperability

Standards matter when you need to integrate with external systems, share models with partners, or ensure your digital twin can evolve with changing technology. Without them, the interoperability of the components of your digital twin, from the sensor communication level, all the way up to running simulations is more challenging.

5.5.1 A framework for standards implementation in your digital twin

Standards in digital twin development aren't a one-size-fits-all solution but exist at different levels, and a successful implementation hinges on prioritizing the right ones for your specific journey. At the foundational level, you'll find communication protocol standards (for example, MQTT and OPC-UA) that govern how data is transmitted between physical and digital assets. Moving up the stack are data model and schema standards (for example, IFC for construction and ISO 15926 for the process industry) which ensure data from disparate sources is semantically consistent and interoperable. Finally, at